



# Predicting Financial Well-Being Using the Financial Capability Perspective: The Roles of Financial Shocks, Income Volatility, Financial Products, and Savings Behaviors

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Accepted: 9 May 2022 / Published online: 8 June 2022

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## Abstract

Efforts to improve family financial well-being usually employ individual-level, knowledge-targeted educational efforts. However, the availability and appropriateness of financial products, fluctuations in the frequency and amount of income (income volatility), the experience of costly unplanned events (financial shocks), and the habits necessary to set aside money for future use (savings behaviors), particularly for those of low income, may play important roles in a family's financial capability and thus well-being. Drawing upon a financial capability framework, this paper assessed structural, interactional, and behavioral predictors' roles in influencing financial well-being. The study used data ( $N=5000+$ ) from the Financial Well-Being Survey and structural equation modeling techniques. Results indicate that saving habit and amount (i.e., behavioral factors) have a far stronger relationship with financial well-being than financial shocks and income volatility and the use of financial products (i.e., structural or interactional factors). These findings suggest that efforts to promote financial capability and thus improved financial well-being should not neglect behaviorally-oriented financial education and intervention programs and policies that encourage savings for families at all income levels.

**Keywords** Financial well-being · Income volatility · Financial shock · Saving behavior · Saving habit · Financial capability

The concept of financial well-being (FWB) has received increased attention in recent years (Collins & Urban, 2019; Kempson et al., 2017; Nanda & Banerjee, 2021; Netemeyer et al., 2018; Prawitz et al., 2006) due in part to the growing focus on individual or family's financial capability and its implications on FWB. To better understand the various components of financial capability and their relationship to FWB, this study draws upon aspects of Sherraden's financial capability model (2013) for exploration and analysis.

Efforts to conceptualize and measure FWB have included the development of several scales and definitions

(Netemeyer et al., 2018; Prawitz et al., 2006; Sunal, 2010). The Consumer Financial Protection Bureau (CFPB, 2017a) developed a working definition of FWB that includes the subjective perception of a person's dual sense of security and freedom of choice, both in the present and future. The CFPB (2017a) then created the Financial Well-Being Scale (FWB Scale) as an index to measure this subjective perception of financial state (see Table 1). Using this scale, the CFPB (2017c) launched the National Financial Well-Being Survey (NFWBS) to understand the demographic characteristics associated with certain levels of FWB in the United States and stimulate research and generate further learning on the topic. For the present study, data from the NFWBS were analyzed to better understand how aspects of Sherraden's financial capability model (2013) measured by financial shocks and income volatility (structural factors), financial product use (interactional factors), and savings behaviors (behavioral factors) predict a household's sense of FWB.

A limited number of studies have empirically explored the pathways of various antecedents to FWB using data from the NFWBS. Most research on this dataset has explored the relationship between behavioral and subjective factors and

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This is one of several papers published together in Journal of Family and Economic Issues on the "Special Issue on the Financial capability and asset building".

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FWB (Lee et al., 2020; Park, 2020; Thornton Walker et al., 2018; Zhao & Zhang, 2020), while inclusion of structural factors, even generally, has been extremely limited (Kabadayi & O'Connor, 2019; Nanda & Banerjee, 2021). Yet, in contemporary theoretical frameworks—including that which the CFPB formulated in its original “Financial Well-Being in America” report (2017),—structures either precede or continually influence individual factors. In these models, behavior is then the final component before FWB (Brüggen et al., 2017; CFPB, 2017b; Sherraden, 2013b). These models frame the person as being influenced by their environment, whereby outcomes are determined by one’s reaction to environmental or structural conditions and the degree of their personal agency under the structural constraints. However, this conceptual pathway was modified in the subsequent 2018 CFPB report (Bureau of Consumer Financial Protection, 2018; Thornton Walker et al., 2018). The model underemphasized the social and economic environment and conceptualized *objective financial situation* as mediating the relationship between individuals’ behavior and FWB. Only access to financial products held was retained as a structural factor within this new category; other structural factors were either procedurally excluded or not considered. This shift in focus is subtle but consequential. It reflects a more conservative perspective that emphasizes personal responsibility for navigating one’s life, dictating the objective financial situations one faces, and securing FWB outcomes. It is worth investigating which conceptualizations have adequate support from current NFWBS data.

As with any study, selecting variables to include in a model and where to place them are partially driven by a need for parsimony, lest the statistical model becomes unwieldy. But it is also driven, either explicitly or implicitly, by the theoretical motivations and perceptions of the authors. For instance, in investigating the drivers of FWB, the Bureau of Consumer Financial Protection (2018) and Thornton Walker et al.’s (2018) modeling reflect an individualist, behavioral-focused perspective in variable selection, while Dorsey et al. (2020) and Nanda and Banerjee (2021) identified variables theoretically related to structural or environmental factors. As such, it is necessary to determine the extent to which key structural and behavioral elements directly predict FWB before mediated pathways and sequencing are presumed.

## Theoretical Framework

This study draws from aspects of Sherraden’s financial capability model (2013) as a guiding framework to examine how certain elements of personal financial capability ultimately lead to FWB. The financial capability model incorporates numerous components that interact in developing financial capability, which, if manifested through actions and

behavior, lead to financial stability, well-being, and development. The social and economic structures set the initial context, starting with the environmental situation in which a household may find itself (Sherraden, 2013). These structural factors entail formal and informal governing processes, norms, and policies generally outside of one’s immediate control, manifesting in varying levels of availability and access to resources, services, and other economic mobility opportunities. Structural factors may include visible opportunities such as availability and access to beneficial financial services and products (Sherraden, 2013) and job availability or less visible but equally consequential factors such as financial socialization (Curran et al., 2018), education (Fernandes et al., 2014), and guidance (Sherraden, 2013). As the theory of planned behavior suggests, people’s structural environments invariably shape their behavior and choices (Fishbein, 2000). Thus, as the structural environments determine what financial products, services, and socialization opportunities are available to individuals and households (Sherraden, 2013), these financial products and socialization then impact one’s ability to act with accurate knowledge and shape opportunities to act (Johnson & Sherraden, 2007; Sherraden, 2013). When combined, the products, socialization, knowledge, and opportunities ultimately lead to financial capability, whereby actions and behaviors are carried out, shaping the perception of FWB (Sherraden, 2013).

Many complex interactions between one’s history, environment, social context, and values create a sense of FWB, as Sherraden’s (2013) and other financial capability models (Storchi & Johnson, 2016) explain. For this study, we first assess the FWB scale’s measurement structure and then draw from Sherraden’s financial capability framework to examine the relative contributions of several antecedents of financial capability. Because of data constraints, we use indicators that reasonably approximate certain structural, interactional, and behavioral aspects of Sherraden’s framework (see Fig. 1). To assess structural predictors, we model concurrently financial shocks and volatile income. While shocks and volatility can sometimes reflect personal choices, we surmise that both indicators are predominantly externally driven and thus fall largely outside the control of the individual. Interactional predictors were modeled with financial products and services, the use of which entails the individual engaging (a behavior) with the range of available products (per the environmental structures). The contributions of shocks, volatility, products, and services have all previously been explored independently (Morduch & Siwicki, 2017; Sherraden, 2013). Behavioral predictors, those elements of the model over which an individual has the greatest control, are assessed by looking at savings habits and the actual amount in savings.

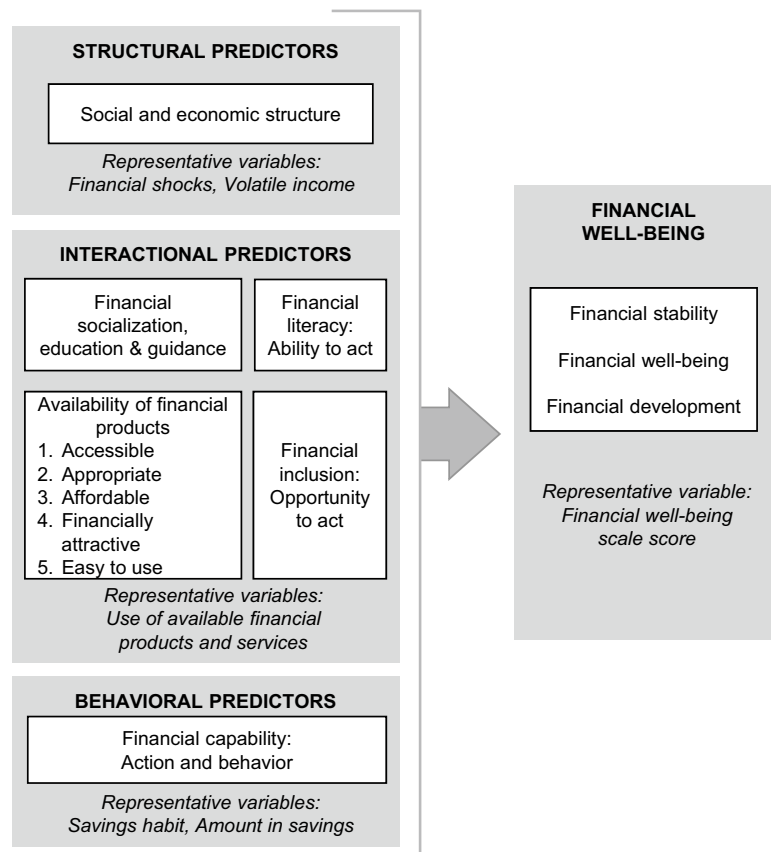
## Financial Shocks and Income Volatility

Financial shocks and income volatility are two important considerations of the financial capability model, which can dramatically impact the availability of financial resources at a given moment. Financial shocks are those events that happen to a person or household, which may induce stress and require adapting to address an externally imposed hardship. Examples of financial shocks could include losing a job, major car repair, or costly medical expenses, and can have a significant impact on FWB, depending on the monetary cost, the disruption to family life, members' perception of it (Asebedo & Wilmarth, 2017), and their ability to address it with other available resources. The U.S. Financial Diaries study highlighted the impact financial shocks have on American households, noting that it can take months or years to recover from the resultant financial predicament (Morduch & Schneider, 2017). One conventional measure of a shock is an unexpected expense that requires, on average, \$2000 to address (Pew Charitable Trusts, 2017). For the 69% of American families who have less than \$1000 in a savings account (Huddleston, 2019), a financial shock can result in cascading effects leading to eviction, job loss, or poorer health due to increased stress, all of which can have additional associated financial challenges.

The 2017 Pew Charitable Trusts report noted that those with a stable income were more likely to come up with the \$2000 to address a shock than those who experienced volatile income flows (2017). Income volatility is considered the fluctuation on a month-to-month basis of one's income (Morduch & Schneider, 2017). Depending on employment availability and employer scheduling, family income will often fluctuate. The severity of those income fluctuations can impede their ability to generate short-term savings, plan for future expenses, meet immediate needs, or address financial shocks (Morduch & Siwicki, 2017; The Aspen Institute, 2019). Beyond the short-term effects, a growing body of research highlights the multidimensional detrimental impact income volatility has on families, including food insecurity (Dahl et al., 2014; Hernandez & Ziolo-Guest, 2009; Wolf & Morrissey, 2017), disruptions in child well-being (Hardy, 2014; Hill et al., 2013; Wolf & Morrissey, 2017), family structure (Mansour, 2018; Nunley & Seals, 2010), and mental health (Corman et al., 2012; Prause et al., 2009).

Because financial shocks and income volatility are generally outside of a family's immediate control, for this study, they are considered examples of structural dynamics and have been shown to pose significant, negative threats to personal financial economy. It is anticipated that data from the NFWBS will show that these vagaries, situated as

**Fig. 1** Conceptual model of structural, interactional, and behavioral predictors of financial well-being. *Note* Authors' conceptualization of predictors are in gray boxes, representative variables are in italics, and components of Sherraden's financial capability model (2013) are portrayed in white boxes



environmental and structural components of financial capability, shape FWB to a substantial degree.

## Financial Products

Financial products fall within what we have termed the interactional aspect of financial capability (Sherraden, 2013), entailing both structural and behavioral considerations in terms of what products and services are available in a given geographic area, and if a person chooses to utilize them. The type of product used, mainstream or alternative, may positively or negatively impact FWB. Mainstream financial products can be understood as gateway tools for building assets and FWB; owning a savings account has been found to predict having an emergency fund (Despard et al., 2020). These products include having checking, savings, life insurance, or retirement accounts. In recognition of the role accounts have in fostering FWB, the Bank On movement aims to connect individuals and families with basic banking accounts (Cities for Financial Empowerment Fund, 2016), while the former myRA program provided a guaranteed growth, no-fee, starter retirement account (United States Department of the Treasury, 2018). These initiatives, and others, such as offering Individual Development Accounts (Sherraden et al., 2015) or Youth Savings Accounts (Ansong et al., 2019), seek to enhance financial stability and thereby FWB by connecting users with affordable and appropriate accounts to generate, hold, or protect assets.

So-called “alternative” financial products and services are used by families who lack access to, choose not to use, or use in addition to the mainstream products due to indebtedness, geographic proximity, educated awareness, ease of access, or creditworthiness. Including payday loans, check cashing services, tax refund anticipation loans, and prepaid debit cards, these alternative products often have high fees and no wealth-building component, but have relatively low to no barriers to access (Karger, 2005; Morduch & Schneider, 2017). Due to the wealth stripping nature of these products (whereby the high-cost fees and interest rates usually carried on the products extract the money that could otherwise be saved), their use is generally considered to harm FWB; it is framed as such in the present study. However, it should be considered that in the absence of mainstream products, the ability of these alternative services to facilitate financial transactions may make their value worthwhile to many who use them (Birkenmaier & Fu, 2016).

## Saving Behavior

The final aspect of building financial capability pertains to behavioral considerations (Sherraden, 2013), including saving, budgeting, and other money management activities mainly within the individual’s control. The strength of

family members’ savings habits and the actual amount in savings may be important markers of an individual’s *ability to behave and act* to leverage available asset accumulation opportunities to save their income and resources. The choice to actively save can translate into a greater accumulation of savings. In other words, the decision to retain or postpone consumption and expenditures through behaviors such as budgeting, bill payment habits, self-control, and long-term goal planning and follow-through (Netemeyer et al., 2018) affect savings amount. Thus, we conceptualize savings habits and savings amount as important behaviorally-driven factors. We must clarify that this conceptualization does not necessarily discount the influence of structural opportunities on either saving habits or the amount saved, including receipt of a tax refund, an inheritance, some other lump-sum receipt event, or even automated savings processes.

In a study of FWB, such as the current study, not only is it crucial to focus on saving habits and the amount because of their direct implications on FWB, but also because of their close relationship to other FWB antecedents, such as the ability to mitigate financial shocks (Despard et al., 2018; Morduch & Schneider, 2017; Stavins, 2020) and weather income volatility (Mills & Amick, 2010; The Aspen Institute, 2019), or use beneficial financial products to hold cash reserves (Despard et al., 2020; The Aspen Institute, 2019). Thus, understanding the relationship between savings habits and amounts and the other variables becomes particularly important as they define the level of FWB.

## Research Objectives

Because FWB has such an important bearing on the overall sense of well-being (Netemeyer et al., 2018), it is important to understand its range of predictors. Thus, this study asks how do financial shocks and income volatility, financial product use, and savings behaviors influence FWB? This question is adapted from a call for proposals by the CFPB (Abt Associates, 2019) and contributes to the body of research on building financial capability for all (Sherraden et al., 2015), as a contemporary societal need. Based on the challenges posed by financial shocks and volatile income, it is essential to consider how much personal behavior accounts for FWB when confronted with forces outside the family’s control. On the other hand, if mainstream financial products and good savings behavior are strong predictors of FWB, it could suggest that personal agency and behaviors are worth strengthening to help families advance their financial well-being. It is hypothesized then that financial shocks and income volatility (structural factors in this study) have statistically observable contributions to FWB.



## Method

### Study Design and Sample Recruitment

The current study used a cross-sectional nationally representative dataset from the (CFPB, 2017c). The CFPB administered the survey to household representatives (age 18+) in the United States through a web-based survey platform. Drawing from aspects of the financial capability model (Sherraden, 2013), a series of variables were then selected for the present study from the dataset as drivers of FWB, namely financial shocks, income volatility, financial product use, and savings behaviors.

Adults aged 18 and older, from all 50 states and Washington, DC, were originally recruited into the KnowledgePanel® by GfK (a German market research company) using address-based sampling by mail (employing a disproportionate stratified sampling by race) and then dual-frame landline and cellphone random digit dialing by telephone (CFPB, 2017c). Recruitment into the NFWBS was conducted in English and Spanish, and willing participants from among the KnowledgePanel group were then enrolled online. Non-internet households were provided with a computer and free internet service to participate in the study, serving as their incentive. Those who used their own technology to access the online panel were enrolled in a points reward program that could be redeemed for cash, merchandise, gift cards, or game entries. The average value of the incentives for the NFWBS (taking 15–25 min) was roughly \$1, and thus not enough to meaningfully influence FWB.

From the KnowledgePanel body, 14,402 panelists were selected, and 5000 surveys were obtained from the general population. An additional sample of 395 surveys from African Americans (non-Hispanic), Hispanics, and those with incomes 200% below the federal poverty line, were obtained, as well as an oversample of 999 participants who were age 62 or older (CFPB, 2017c). This produced a total sample size of  $N=6394$ , which constituted a nationally representative, quota sampled panel with target demographics matching the prior U.S. census. For the present study, only those who answered the questions on volatility ( $n=6330$ ), savings habits ( $n=6374$ ), and amount in savings ( $n=5210$ ) were used, reducing the analytic sample to  $n=5176$ .

### Measures

#### Outcome Variable

The outcome variable, *FWB*, was measured using a 10-item FWB scale on a 5-point response scale. The CFPB developed the index to measure one's present and future perception of choice and security pertaining to their FWB

(CFPB, 2017a). The 10 items measure how well and often a given statement applies to respondents or their situation (see Table 1). Higher scores indicate greater FWB (CFPB, 2017a). The CFPB measured the marginal reliability statistic for the scale when administered online with adults aged 18 to 61 (0.89) and 62+ years (0.90), and it produced values comparable to high Cronbach's alphas, suggesting that the scale has high reliability (CFPB, 2017a).

#### Predictor Variables

*Financial shocks* were measured using a cumulatively scored 11-item binary response index ranging from 0 to 11. Shocks were defined as incidents that occurred within the last 12 months that had significant financial implications, including loss of a job, reduced work hours and/or pay from an employer or family-owned business, receipt of a foreclosure notice, a major car or home repair, a health emergency, and a divorce or separation (CFPB, 2017c). Other shocks included a child added to the household, the death of a primary breadwinner, receipt of a large sum of money beyond normal income (such as inheritance, bonus, or other windfalls), a child started daycare or college, and the provision of unexpected financial support to a family member or friend. While these shocks may not all be equal in financial impact, the perception of distress could independently vary, depending on family resources and resilience. According to the family stress theory (Hill, 1949), the presence of shocks and the ability to handle them are more important in subjective terms than the objective dollar amount the shock incurs. Thus, a cumulative index was used to measure the presence of financial shocks.

*Income volatility* was measured with a single question (CFPB, 2017c). Respondents were asked to describe the severity of the month-to-month changes in their income using a 3-point response set: 1 (*roughly the same each month*), 2 (*roughly the same most months, but some unusually high or low months during the year*), or 3 (*often varies quite a bit from one month to the next*). In this study, responses 2 and 3 were collapsed to form a binary variable coded as 1 (*yes*) and 0 (*no*).

*Financial products* held were operationalized using a cumulatively scored 8-item, binary response index of consumer products, ranging from 0 to 8. Products held included checking or savings accounts at a bank or credit union, life insurance, health insurance, a retirement account (such as a 401k or IRA), a pension, non-retirement investments (such as stocks, bonds, or mutual funds), an education savings account, and a student/education loan for one's self or someone else (CFPB, 2017c).

*Alternative financial products* used by consumers were operationalized using a cumulatively scored 5-item, binary response. Products used within the prior 12 months

included a payday loan or cash advance loan, a pawn loan or auto title loan, a reloadable card that was not linked with a checking or savings account, a place other than a bank or credit union to give or send money to relatives or friends outside the United States and a place other than a bank or credit union to cash a check or purchase a money order (CFPB, 2017c). For statistical modeling purposes, the summed scores were dichotomized into 1 (yes) for the use of any alternative product and 0 (no) for those who used none of the products.

*Savings habit* was measured with one question. Participants were asked to describe the extent to which they agreed with the idea that putting money into savings was a habit for them (CFPB, 2017c). Responses ranged from 1 (*strongly disagree*) to 6 (*strongly agree*).

*Savings amount* was measured according to respondents' self-report of the amount of money they had in savings (in cash, checking, and savings account balances combined) at the time of data collection. Respondents choose from 13 response options: \$0, \$1 to \$49, \$50 to \$99, \$100 to \$249, \$250 to \$499, \$500 to \$999, \$1000 to \$1999, \$2000 to \$4999, \$5000 to \$9999, \$10,000 to \$19,999, \$20,000 to \$49,999, \$50,000 to \$74,999, and \$75,000 or more. Those who responded "I don't know" or "Prefer not to say" were coded as missing and excluded from analysis (CFPB, 2017c).

### Control Variables

Several demographic characteristics, particularly related to wealth disparities, are recognized to influence financial status, including race and ethnicity (due to racism; Asante-Muhammad et al., 2017), age (Collins & Urban, 2019), gender (Chang, 2010), and household income levels (Netemeyer et al., 2018). Key control variables were collected during the initial enrollment into the KnowledgePanel. In this paper, the race and ethnicity options were collapsed into a single measure of whether participants self-identified as Black non-Hispanic, coded as 1 (*yes*) and 0 (*no*). Age was originally measured as an ordinal variable, with participants indicating their age group: "18–24", "25–34", "35–44", "45–54", "55–61", "62–69", "70–74", and "75+". For this paper, the age groups were collapsed into one binary variable, indicating whether participants were 62 years and older (coded as 1) or not (coded as 0). This binary categorization aligns with how the CFPB accounts for age in the usage of the FWB scale. Gender was measured as a binary variable and coded 1 (*female*) and 0 (*male*). Household income was measured in terms of income range with nine response options and coded as 1 (*less than \$20,000*), 2 (*\$20,000 to \$29,999*), 3 (*\$30,000 to \$39,999*), 4 (*\$40,000 to \$49,999*), 5 (*\$50,000 to \$59,999*), 6 (*\$60,000 to \$74,999*), 7 (*\$75,000 to \$99,999*), 8 (*\$100,000 to \$149,999*), and 9 (*\$150,000 or more*). In this paper, the

income variable was dichotomized (Kahneman & Deaton, 2010) as high income, coded as 1 (*earns \$75,000 or more*) and 0 (*earns less than \$75,000*). The analysis also controlled for participants' poverty status, coded 1 (*within 200% above the federal poverty line*) and 0 (*no*).

### Data Analysis

We used a structural equation modeling (SEM) approach with Mplus 8.4 to address the hypotheses. SEM is ideal for accounting for measurement errors while modeling relationships among observed and latent variables (Kline, 2015). We used a sequential modeling approach, starting with the measurement model, a reduced structural model (without covariates), and a full model (with covariates). The CFPB provides a scoring mechanism for creating a single composite score from the 10 FWB items. However, when the individual items in a scale have different distributions, the combined score can be prone to errors. A thorough examination of the 10 CFPB items' distributions revealed uneven distributions, ranging from negatively skewed to normally distributed to positively skewed items. Thus, we used confirmatory factor analysis (CFA) to verify the FWB factor structure.

After confirming a good measurement model, we tested the structural relationships between six predictors (structural, interactional, and behavioral constructs) and FWB in three separate models. The first was a *reduced SEM model*, which focused on the predictive role of the three construct factors. The second was a full SEM model that expanded the reduced SEM model by accounting for five covariates (age, gender, race, income, and poverty status). The third model—the *mediation model*—expanded the full mediation model by testing an indirect pathway from saving habit through saving amount to FWB.

As shown in Table 2, several predictors and covariates vary by the census division. Therefore, we used the CLUSTER option in Mplus to account for potential clustering at the census divisional level in all models. Sampling weights were also applied in all modeling to retain the national representativeness of the sample. We applied the weighted least square mean and variance adjusted (WLSMV) estimator to all models because of the predictors and covariates' binary and ordinal nature. The following model fit indices were used to assess the adequacy of all measurement and structural models: CFI and TLI > 0.90, and RMSEA < 0.05 with a 90% CI < 0.08 upper bound (Hu & Bentler, 1999).

## Results

### Sample Description

Of the sampled group ( $n = 5176$ ), 52% identified as male, and 47% identified as female. Those who identified as non-Hispanic categorized themselves as White (70%), Black (10%), and Other (5%), whereas those who identified as Hispanic comprised 13%, with the remaining unidentified or mixed. Those aged 62 and older comprised 35% of the sample, while the remaining 65% were 18 to 61. Most respondents reported annual household income levels of at least 200% or more above the poverty level (76%); 45% reported an income of \$75,000 or more. Table 2 presents the descriptive characteristics for the full sample as well as the nine census divisions.

### Measurement Model

The initial FWB measurement model shown in Table 2, without the correlated errors, produced a poor fit ( $\chi^2 = 1749.751$ ,  $df = 35$ ,  $p < 0.001$ ;  $CFI = 0.878$ ;  $TLI = 0.843$ ;  $RMSEA = 0.088$  [90% CI 0.084–0.091]). Based on suggested modification indices, six correlated errors were incorporated into the FWB measurement model, which led to a good model fit ( $\chi^2 = 522.915$ ,  $p < 0.001$ ;  $CFI = 0.965$ ;  $TLI = 0.945$ ;  $RMSEA = 0.052$  [90% CI 0.048–0.056]). As shown in Table 1, all the standardized factor loadings were significant at the 0.001 level and ranged from 0.597 to 0.785, above the 0.30 threshold.

### Structural Model

The contributions of the aspects of behavioral and structural factors tested in this study as predictors of FWB are illustrated in Fig. 2. The structural models without covariate (reduced SEM model:  $\chi^2 = 1,279.373$ ,  $df = 83$ ,  $CFI = 0.937$ ,  $TLI = 0.921$ ,  $RMSEA = 0.053$  [90% CI 0.053, 0.055]) and with covariates (full SEM model:  $\chi^2 = 1761.856$ ,  $df = 128$ ,  $CFI = 0.926$ ,  $TLI = 0.911$ ,  $RMSEA = 0.050$  [90% CI 0.048, 0.052]) both had good fit to the data. However, a chi-square difference test (see Table 3) suggested that the full model that controlled for covariates produced significant improvement over the reduced model ( $\chi^2 = 482.483$ ,  $df = 45$ ,  $p < 0.001$ ). As shown in Fig. 2, all the standardized factor loadings for the FWB scale remained high in the final structural model, ranging from 0.624 to 0.824.

A third structural model that tested possible mediations produced poor fit to the data (mediation model:  $\chi^2 = 3,605.983$ ,  $df = 137$ ,  $CFI = 0.874$ ,  $TLI = 0.848$ ,  $RMSEA = 0.063$  [90% CI 0.062, 0.065]). The full model was therefore retained as the final model. The standardized path estimates showed that all but one predictor was statistically significant (Alternative financial products:  $\beta = -0.025$ ,  $p = 0.212$ ). All the model fit indices are presented in Table 3.

The findings also show that the main behavioral factors are positive predictors of FWB as expected ( $\beta_{\text{Saving amount}} = 0.437$ ,  $SE = 0.020$ ,  $p < 0.001$ ;  $\beta_{\text{Saving habit}} = 0.311$ ,  $SE = 0.017$ ,  $p < 0.001$ ). When savings amount increases by one standard deviation, FWB increases by 0.437 standard deviations. Also, FWB increases by 0.311 standard deviations for every one standard deviation increase in saving habits. The two main structural factors are negatively associated with FWB as expected ( $\beta_{\text{Financial shock}} = -0.086$ ,  $SE = 0.009$ ,  $p < 0.001$ ;  $\beta_{\text{Volatile income}} = -0.025$ ,  $SE = 0.011$ ,  $p < 0.01$ ). Thus, FWB

**Table 1** CFPB financial well-being items and factor loadings from the measurement model

| Question stem; response set                                                                                                                                                                                     | Items                                                                                                             | Standardized factor loadings |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|------------------------------|
| How well does this statement describe you or your situation?<br>• Describes me completely<br>• Describes me very well<br>• Describes me somewhat<br>• Describes me very little<br>• Does not describe me at all | P1.1. I could handle a major unexpected expense                                                                   | 0.715                        |
|                                                                                                                                                                                                                 | P1.2. I am securing my financial future                                                                           | 0.648                        |
|                                                                                                                                                                                                                 | P1.3. Because of my money situation, I feel like I will never have things I want in life (R)                      | 0.707                        |
|                                                                                                                                                                                                                 | P1.4. I can enjoy life because of the way I'm managing my money                                                   | 0.672                        |
|                                                                                                                                                                                                                 | P1.5. I am just getting by financially (R)                                                                        | 0.597                        |
|                                                                                                                                                                                                                 | P1.6. I am concerned that the money I have or will have won't last (R)                                            | 0.622                        |
| How often does this statement apply to you?<br>• Always<br>• Often<br>• Sometimes<br>• Rarely<br>• Never                                                                                                        | P2.1. Giving a gift for a wedding, birthday or other occasion would put a strain on my finances for the month (R) | 0.825                        |
|                                                                                                                                                                                                                 | P2.2. I have money left over at the end of the month                                                              | 0.785                        |
|                                                                                                                                                                                                                 | P2.3. I am behind with my finances (R)                                                                            | 0.721                        |
|                                                                                                                                                                                                                 | P2.4. My finances control my life (R)                                                                             | 0.724                        |

Measurement-only model fit:  $\chi^2 = 3651.456$ ,  $p < 0.001$ ;  $CFI = 0.973$ ;  $TLI = 0.962$ ;  $RMSEA = 0.049$  (0.047–0.050). \*\*\* $p < 0.001$ ; (R): Reverse coded items; All loadings are statistically significant at the 0.001 significance level

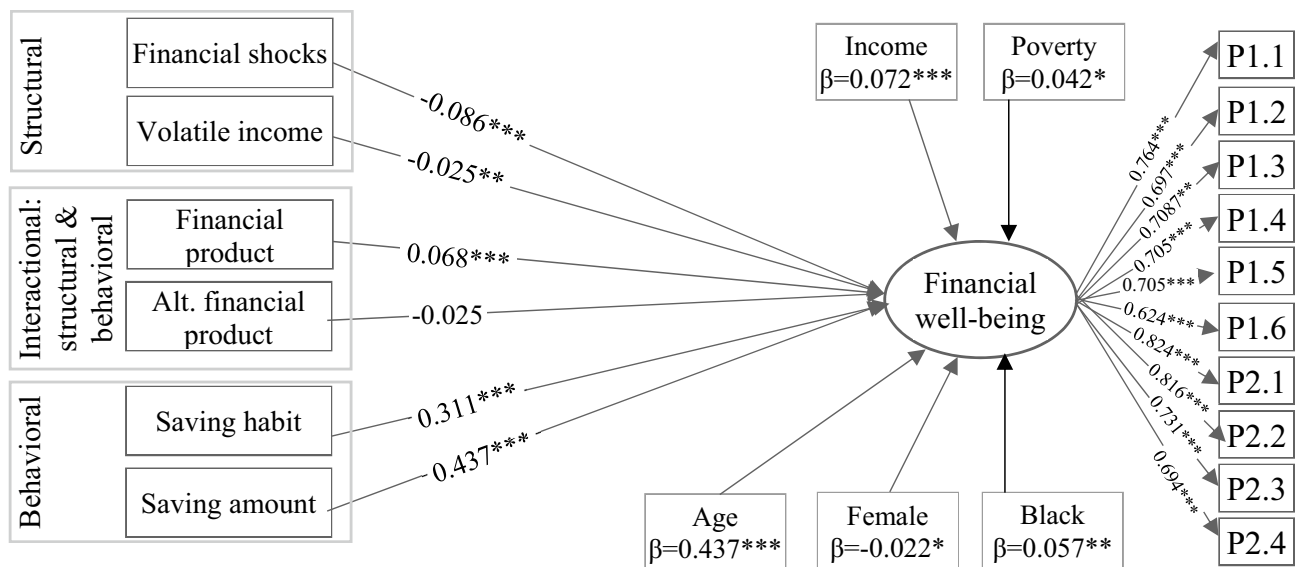
**Table 2** Summary statistics of the full sample and broken down by the nine census divisions

|                                   | Census division |             |              |                    |                    |                |                    |                    |            |            |
|-----------------------------------|-----------------|-------------|--------------|--------------------|--------------------|----------------|--------------------|--------------------|------------|------------|
|                                   | Full sample     | New England | Mid-Atlantic | East-North Central | West-North Central | South Atlantic | East-South Central | West-South Central | Mountain   | Pacific    |
|                                   | Mean(SD)        | Mean(SD)    | Mean(SD)     | Mean(SD)           | Mean(SD)           | Mean(SD)       | Mean(SD)           | Mean(SD)           | Mean(SD)   | Mean(SD)   |
| Savings amount (range)***         | 4.48(1.80)      | 4.66(1.76)  | 4.54(1.78)   | 4.65(1.75)         | 4.52(1.72)         | 4.56(1.82)     | 4.43(1.87)         | 4.27(1.88)         | 4.34(1.77) | 4.67(1.80) |
| Saving habit                      | 4.38(1.45)      | 4.31(1.54)  | 4.36(1.49)   | 4.30(1.54)         | 4.40(1.47)         | 4.44(1.43)     | 4.40(1.48)         | 4.37(1.45)         | 4.36(1.46) | 4.45(1.41) |
| Access to mainstream products***  | 3.51(1.79)      | 3.67(1.80)  | 3.63(1.79)   | 3.67(1.80)         | 3.88(1.70)         | 3.54(1.79)     | 3.53(1.83)         | 3.18(1.86)         | 3.60(1.71) | 3.28(1.81) |
| Financial shocks**                | 0.78(1.01)      | 0.72(0.95)  | 0.73(0.97)   | 0.72(0.95)         | 0.73(0.98)         | 0.82(1.02)     | 0.79(0.95)         | 0.87(1.08)         | 0.89(1.22) | 0.80(0.98) |
|                                   | %               | %           | %            | %                  | %                  | %              | %                  | %                  | %          | %          |
| Income volatility (1 = Yes)       | 27              | 26.1        | 28.74        | 26.18              | 26.75              | 28.1           | 26.55              | 26.47              | 24.19      | 27.88      |
| Alt fin. Products (1 = Yes)***    | 20              | 11.8        | 19.76        | 19.08              | 17.6               | 21.75          | 25.32              | 27.57              | 15.35      | 20.38      |
| 200% + poverty level (1 = Yes)*** | 76              | 80.74       | 77.5         | 76.22              | 82.82              | 77.54          | 75.32              | 66.62              | 75.05      | 76.08      |
| Age 62 years + (1 = Yes)*         | 35              | 34          | 34           | 34                 | 35                 | 38             | 37                 | 31                 | 36         | 34         |
| Income \$75,000+ (1 = Yes)***     | 45              | 55.27       | 48.92        | 42.35              | 47.6               | 45.6           | 40.9               | 39.66              | 40.94      | 50.91      |
| Female (0 = Male)                 | 47              | 47.51       | 50.35        | 48.35              | 49.78              | 46.37          | 50.64              | 48.24              | 44.78      | 45.07      |
| Race***                           |                 |             |              |                    |                    |                |                    |                    |            |            |
| White non-Hispanic                | 70              | 84.78       | 71.78        | 80.21              | 90.43              | 67.66          | 74.35              | 53.14              | 75.279     | 57.13      |
| Black non-Hispanic                | 10              | 4.04        | 12.97        | 10.98              | 3.04               | 18.13          | 21.1               | 10.72              | 2.56       | 5.45       |
| Hispanic                          | 13              | 7.45        | 11.19        | 5.49               | 3.92               | 10.87          | 1.95               | 30.17              | 15.57      | 25.55      |
| Other non-Hispanic                | 5               | 3.726       | 4.05         | 3.29               | 2.61               | 3.32           | 2.597              | 5.97               | 6.61       | 11.87      |

Alt fin. products alternative financial products; SD standard deviation

\* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$





**Fig. 2** Results of Behavioral, Interactional, and Structural Predictors of the CFPB Financial Well-Being Scale. *Note* The following correlated errors, all significant at the 0.001 significance level, are not

shown in the model: P1.2  $\Leftrightarrow$  P1.4 ( $\beta=0.398$ ), P1.1  $\Leftrightarrow$  P1.4 ( $\beta=0.316$ ), P1.1  $\Leftrightarrow$  P1.2 ( $\beta=0.314$ ), P1.3  $\Leftrightarrow$  P1.6 ( $\beta=0.270$ ), and P2.2  $\Leftrightarrow$  P2.4 ( $\beta=-0.151$ )

**Table 3** Model fit indices of tested models and chi-square difference test results

| Models                                                    | N     | $\chi^2(df)^{***}$ | CFI   | TLI   | RMSEA [90% CI]       | $\Delta\chi^2(df)$ |
|-----------------------------------------------------------|-------|--------------------|-------|-------|----------------------|--------------------|
| 1. Measurement model (i.e., without correlated errors)    | 6,389 | 1749.751(35)       | 0.878 | 0.843 | 0.088 [0.084, 0.091] |                    |
| 2. Final measurement model (i.e., with correlated errors) | 6,389 | 522.915(29)        | 0.965 | 0.945 | 0.052 [0.048, 0.056] | 1226.836(6)***     |
| 3. Reduced SEM model (i.e., without covariates)           | 5,175 | 1279.373(83)       | 0.937 | 0.921 | 0.053 [0.053, 0.055] |                    |
| 4. Full SEM model (i.e., with covariates)                 | 5,175 | 1,761.856(128)     | 0.926 | 0.911 | 0.050 [0.048, 0.052] | 482.483(45)***     |
| 5. Mediation model                                        | 6,320 | 3,605.983(137)     | 0.874 | 0.848 | 0.063 [0.062, 0.065] |                    |

\* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$

decreases by 0.086 and 0.025 standard deviation units for every standard deviation increase in financial shocks and income volatility, respectively. Regarding the relative magnitude of the relationships, the behavioral factors related to saving ( $\beta=0.437$  to  $0.311$ ) are more predictive of FWB than any of the structural factors ( $\beta=-0.086$  to  $-0.025$ ).

Of the two types of financial products—usage of mainstream and alternative financial products—only mainstream financial products were significantly positively related to FWB ( $\beta=0.068$ ,  $p < 0.001$ ). That means greater use of such mainstream products is associated with a higher sense of FWB. On the other hand, using alternative financial products is negatively associated with FWB ( $\beta=-0.025$ ), but that relationship is not statistically significant at the 0.05 level.

## Discussion and Implications

This exploratory analysis sought to determine how financial shocks, income volatility, financial product use, and savings behaviors contribute to FWB. Beyond this, it was

hypothesized that the variables related to structural and environmental conditions—financial shocks and income volatility—are more predictive of FWB, than the other two variable groups over which the family has more control. The current NFWBS data did not support this hypothesis. Rather, evidence from the data suggests that savings behaviors, the variable most within respondents' control are more predictive of FWB than hypothesized. This finding initially suggests that some indicators of the financial capability model—particularly financial shocks, income volatility, financial products, alternative financial products, saving habit, and saving amount—may not always contribute equally to FWB. If experimental and replication studies arrive at a similar conclusion, it will lend credence to a more individually focused conception of financial capability development. Further confirmation would mean some strategic intervention areas may need more attention to yield greater impact on FWB. Thus, we strongly encourage replication of our study with different datasets from different contexts.

Regarding the direction of the relationships between the variables and FWB, the results trended in the expected direction. Controlling for household income, poverty status, age, and race, we found that financial products, saving behaviors, and actual amounts saved all positively contributed to FWB. It is also worth noting that among all the predictors of interest and covariates, saving habit and amount have the strongest contribution to FWB per the NFWBS data. Conceptually, this makes sense: family members earn money, which goes into a bank account. Consciously saving a portion of the money serves as a cushion that provides a sense of security and choice, both now and in the future, which is the definition of FWB. Financial shocks, volatile income, and costly alternative financial products, which were hypothesized to predict FWB negatively, indeed may have adverse implications for FWB, as illustrated in the model. However, while these may have a statistically detectable impact on the respondents' well-being, overall, they had weak contributions to FWB perceptions.

### Financial Shocks and Income Volatility

While financial shocks and income volatility negatively predict FWB, the relationship was much smaller than expected. This finding from the NFWBS data is contrary to Morduch and Schneider's findings (2017) on the dynamics and impact of income volatility and financial shocks, where these features of personal economy shape perceptions of life, restrict "mental bandwidth" and foster scarcity mindsets (Mullainathan & Shafir, 2013), and drastically impact the sense of security and choice available to the family. The discrepancy could be due to the present study's sampling of a much lower proportion of low- and middle-income earners than those who comprised the U.S. Financial Diaries Study (Morduch & Schneider, 2017). According to Netemeyer et al. (2018), lower-income families would perceive the effects of income volatility and shocks more severely than those of higher income levels, whereby income itself can mitigate the severity of fluctuations in income and expenses. The sample used in the present study includes those of higher incomes, with 45% of the sample earning \$75,000 or more, an average threshold level for meeting basic needs and achieving emotional well-being (Kahneman & Deaton, 2010).

Future research should aim to better understand the relationship's dynamics by testing the moderating role of income levels in the relationship between income volatility and financial shocks and FWB. The current study could not explore these dynamics because of poor model fit to the NFWBS data. However, insights from further analyses could impact how financial education is taught, and financial products are designed. Certain budgeting and money management strategies that work for those at higher incomes—and most likely those designing the curriculum—may not be as

effective or applicable in addressing low-income economic realities, such as the need to generate and access short-term savings in an environment of volatile income and expenses. The financial capability model underscores the need for appropriate products and services (Sherraden, 2013), which often are designed to be universal in application, but end up disproportionately penalizing families with lower incomes who have smaller savings buffers against financial shocks. Different income levels may then warrant different designs, or at least a universally basic design, such as "safe accounts" advocated by the Bank On movement (Cities for Financial Empowerment Fund, 2016).

### Financial Products

The average respondent held at least three mainstream financial products, while only 20% used one or more alternative products in the past 12 months. Due to indexing, a nuanced analysis was not conducted to determine which products were used. However, it could be possible that because savings was a strong predictor, those products associated with savings (checking, savings, retirement, education, and investment accounts) would have a mediating relationship between savings and FWB, while only accounting for, on their own, a small amount of the relationship. Again, having appropriate financial products and services available to all income levels would be important in supporting FWB for the entire population.

### Savings Behaviors

Money management behaviors measured by savings habits and amount in savings were the largest predictors of interest ( $\beta = 0.311, 0.437$ ), a finding supported by the CFPB's reports (Bureau of Consumer Financial Protection, 2018; Thornton Walker et al., 2018). Research suggests that while savings depend on income, the ability to save is not exclusively the domain of the middle- or high-income earner (Sherraden & McBride, 2010). Rather, combined with the appropriate savings vehicles and the regular act of savings, accumulating a financial cushion is possible for most families. While the relationships between different amounts in savings and FWB were not explored, having the belief that saving is possible and the agency to act on that belief could comprise a meaningful proportion of the relationship. This is an important distinction: Savings is considered a technical activity increasingly facilitated by interventions informed by the field of behavioral economics (such as auto-enrollments, automatic savings transfers, and opt-out program designs), and traditionally taught as a technical exercise (including lessons on account types, savings rates, and compound interest). Yet,

at its most fundamental, saving is a behavioral endeavor. Saving requires clarity of values and goals, willingness to sacrifice short-term gratification for future expenses, ability to withstand materialistic social forces, and the skill to account for income and ensure that expenses remain below the family's means. Saving money, or as is often done by those with lower income levels, saving tax refunds or stockpiling goods (Morduch & Schneider, 2017), requires intentional behavior. Therefore, developing long-term savings habits takes more than a single class (Fernandes et al., 2014), but like most behavior modification interventions, it requires a sustained, supported, and systematic approach. Financial coaching as a model for helping participants save money and generally develop financial capability is gaining popularity (Collins et al., 2007, 2013; Theodos et al., 2018) precisely because it provides a sustained, supported, and systematic approach. But because coaching is resource intensive, usually conducted one-on-one and over a longer period, other behavioral change models focused on saving may be worth exploring and developing.

## Demographics

Beyond the main variables under study, demographics contributed modestly to the model, supporting other's findings on race and ethnicity, gender, and household income (Asante-Muhammad et al., 2017; Chang, 2010; Collins & Urban, 2019; Netemeyer et al., 2018), while age was the strongest covariate ( $\beta = 0.437$ ). These demographic and socioeconomic factors are largely outside the scope of a family's control and yet contribute in a meaningful way to FWB. These factors are often the reason behind differential treatment, such as receiving lower wages due to bias or locating bank branches outside ZIP codes with lower median incomes. In other words, these variables may reflect the impact structural advantages and disadvantages have on family well-being. However, this study did not dive into such structural nuances because data collected for the NFWBS generally focused on individualistic approaches to money management (CFPB, 2017d). Future research needs to be conducted to discern any differences in how money is managed from a cultural context. Many minority groups engage in collective approaches to save money and support kin. Such interdependent approaches can include savings circles, lending money to family, or outright financially supporting adult children or parents (Morduch & Schneider, 2017; Stack, 1974).

Additionally, recent literature has explored differences in what people save according to income type (Morduch & Siwicki, 2017). Those with steady paychecks can often benefit from automatic savings mechanisms to save small amounts periodically. However, those with low and/or

volatile income, who often have smaller financial shock buffers, may save only on occasion, such as when bonuses or tax refunds arrive, or save goods, stockpiling when they cost less. Saving and then spending in short cycles of a few months to smooth the impact of income and expense volatility is another example of how savings differ by income type (Morduch & Schneider, 2017). Assessing different ways of saving according to income level and determining how this impacts FWB is another area for future inquiry.

## Generational Age

The relation between FWB and age also warrants mention due to its predictive strength ( $\beta = 0.437$ ). Contemporary economic and life stage conditions are such that Millennials, early in life and career, often face high debt and expenses, especially if they hold student loans and are starting families. This conceptually can result in lower levels of FWB. Compared to the Millennial group, it would be expected that the Gen X group would show a slight improvement in FWB, having higher earnings and lower childcare expenses. However, the Boomer and Pre-Boomer generations have likely paid off most student loan and mortgage debt, have lower expenses now that their children are not being supported, and have accumulated savings for use in retirement. Most in these oldest cohorts are also drawing down on government assistance programs such as Social Security and Medicare. As such, stage in the life course becomes a significant predictor of FWB. It is conceivable that this elder group, who are the most financially secure, holds a more resigned perception of their financial state, and with that acceptance, makes do with the resources they have. Clearly, there are exceptions, particularly among sub-demographic groups. However, as a cohort, these initial findings support a general trajectory of life course improvement in FWB, as was also found by the CFPB (2017b). The degree to which programming and policies can support savings earlier in the life course can potentially have two-fold ramifications. First, savings would enable those in the Millennial and Gen X cohorts to have improved FWB in the face of more challenging economic realities than those faced by the older generations. Also, they could have improved FWB later due to time as a factor in compounding interest on savings over the long term. Currently, those aged 50 and over are eligible for extra "catch-up" contributions to tax favorable IRAs (Internal Revenue Service, 2018). Providing a "running-start" contribution for younger generations could have an enhancing positive impact throughout the life course.

## Limitations and Strengths

The study had several notable limitations. The measure of income volatility, one of the key variables, was based upon the respondents' perceptions, with only three options from which to choose. Recent studies analyzing income volatility and its effects utilize a much more nuanced approach that measures the percent of change from one month to the next, rather than subjecting the response to recall bias (Dahl et al., 2014; Mansour, 2018; Morduch & Siwicki, 2017; Nunley & Seals, 2010; Prause et al., 2009). This method requires a longitudinal study design, which is more appropriate for measuring change over time and establishing causation. The NFWBS dataset was cross-sectional, limiting its ability to assess volatility according to the more commonly accepted method. Also, many financial behaviors comprise money management behaviors, whereas this study only looked at savings. Habits related to budgeting, bill and credit card payment, and activities supporting frugality (CFPB, 2017d) may also contribute to FWB, beyond saving, and thus warrant further exploration. Lastly, it is vital to understand which financial shocks have the greatest impact on FWB, as different shocks would require different practices, policies, or products to mitigate their influence.

Notwithstanding the limitations, the data utilized in this study were nationally representative, and the analysis sufficiently powered, thus supporting the generalizability of the findings. Although a more nuanced study is needed to understand the role that particular types of shocks, financial products, or saving have in predicting FWB, the overall findings contribute to a preliminary empirically-based understanding of financial capability and adds to the body of knowledge relating to how structural and behavioral factors predict FWB.

## Conclusion

Many factors determine one's FWB. Depending on the theoretical orientation, they can result from socioeconomic conditions, the type of financial socialization one received during formative years, and educational opportunities. They could also stem from skills, knowledge, and behaviors. Regardless, it is agreed that access to appropriate financial products and services and the ability and opportunity to act on one's own behalf enhance financial capability levels. Using variables from the NFWBS dataset as proxies for structural, interactional, and behavioral domains, this study explored the relative contributing role of the domains to FWB. The project also adds to the limited number of empirical studies (Çera et al., 2021; McGarity et al., 2020; Sun et al., 2022) that test Sherraden's financial capability model (2013). Of these, it is the first known to empirically incorporate concepts aligned with environmental or structural

influences on FWB and to use the increasingly utilized CFPB's (2017a) FWB Scale. Other ways of conceptualizing and studying the model could be explored in future research. Although it was anticipated that financial shocks and income volatility would be significantly stronger predictors of FWB than financial products or savings behaviors, the findings suggested otherwise. At least for this national population sample, savings behavior variables appeared to be much more influential than shocks, volatility, or financial products. While leading to new questions, these findings support the need for a shift in financial education toward teaching behaviorally oriented approaches to saving and calling for policies that actively support savings for all income levels.

**Author Contributions** Conceptualization, data identification, and writing the first draft of the manuscript was performed by JA-C. DA conducted the data analysis and drafted the results section. Both authors edited previous versions of the manuscript. Both authors read and approved the final manuscript.

**Funding** Jeffrey Anvari-Clark received a nominal honorarium from Washington University in St. Louis for authoring the paper as part of the 5th Financial Capability and Asset Building Convening (September, 2020). David Ansong declined to receive funding for conducting this study.

**Data Availability** The paper uses data obtained from the Consumer Financial Protection Bureau (CFPB, 2017c). The citation in the main article has the full URL where the dataset and associated documentation can be obtained.

**Code Availability** *Mplus* 8.4 was used for calculating data analysis.

## Declarations

**Conflict of interest** The authors have no relevant financial or non-financial interests to disclose.

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